Educ Stud Math (2008) 67:237–254
DOI 10.1007/s10649-007-9100-6

Investigating the secondary–tertiary transition

Ghislaine Gueudet

Published online: 17 January 2008
© Springer Science + Business Media B.V. 2007

Abstract The secondary–tertiary transition has been studied in a great amount of research in mathematics education, adopting different focuses and theoretical approaches. I present here how these focuses led the authors to identify and study different students' difficulties and to develop different means of didactical action. Individual, social, but also institutional phenomena are considered with different perspectives. Each perspective yields a particular view of transition. The association and comparison of these views makes it possible to build an organized outline of this complex object, combining several kinds of ruptures and long-term evolutions.

Keywords Advanced mathematical thinking · Assessment · Institutions · Mathematical language · Mathematical reasoning · Proof · Secondary–tertiary transition

Mathematicians, and mathematics educators, often mention transition from secondary to tertiary level as a major issue. The object they refer to remains nevertheless vague. Our aim in this paper is to display how mathematics education research makes it possible to understand and describe this object.

This paper's origin is a course given at the 13th French mathematics didactics summer school. This course consisted of a literature review about secondary–tertiary transition for scientific students. These students can have different professional projects, such as working in physics, chemistry, computer science, or becoming an engineer or a teacher. But all of them are enrolled in university courses which might be very general but are the courses followed by the students who will eventually become professional

G. Gueudet
Centre de Recherche sur l'Education les Apprentissages et la Didactique (CREAD),
Rennes, France

G. Gueudet (✉)
Institut Universitaire de Formation des Maîtres (IUFM) de Bretagne, 153,
rue Saint-Malo, 35043 Rennes Cedex, France
e-mail: Ghislaine.Gueudet@bretagne.iufm.fr

mathematicians (the transition between secondary and post-secondary courses for other students was addressed in the same summer school in a specific course by Straesser 2007). Because of the amount of literature published since the course was taught, I complemented the summer school course with references enriching the panel of lenses providing insight on transition issues.

1 Introduction

Reviewing the mathematical education literature about secondary–tertiary transition is a challenging task. When does the "transition" happen? There is no straightforward answer to this question; in addition to the different national contexts to consider, different time scales can intervene. In some countries "bridging courses" (Wood 2001) are designed to help students before they enter university, or during their first year. Studying bridging courses is naturally relevant for transition. But transition certainly happens also outside of the period starting at the end of secondary school and finishing at the beginning of university.

Sometimes mathematical topics considered as significantly difficult are proposed only during the second year of university. Sometimes the opposite occurs: mathematical content that could belong to the university's curriculum is already taught at secondary school, in order to prepare students. Significant ruptures can happen at different moments. For this reason, all the educational research concerning a period of time between two years before entering university and two years after is likely to establish results about transition issues.

I have chosen to select, first of all, literature which explicitly addresses transition. It means here that the word "transition" is used by the author as a term for the secondary–tertiary transition or other transition phenomena which might occur in particular when entering university: transition from one mode of thinking to another, transition to formal proof, etc. I try to identify the most striking results in these papers, and to emphasize in particular among the phenomena observed those presented as specific of the secondary–tertiary transition. When such results are significantly connected with results of other research, not explicitly addressing transition, I mention the corresponding works. I am fully aware of the incompleteness of this process; I am certainly leaving aside many of the insightful works published on tertiary education in particular. However it should permit the inclusion of the main research results on transition.

In a first loose approach, the questions related with transition issues can be classified into two categories: observation and analysis of students' difficulties; and proposal and evaluation of teaching designs. But many kinds of difficulties occur, and many hypotheses on their sources can be formulated. For example, in a paper by Moore (1994), entitled "Making the transition to formal proof", seven kinds of sources for the novice students' difficulties with proof are observed:

> The students did not know the definitions, that is, they were unable to state the definitions;
> The students had little intuitive understanding of the concepts;
> The students' concept images were inadequate for doing the proofs;
> The students were unable, or unwilling, to generate and use their own examples;
> The students did not know how to use definitions to obtain the overall structure of proofs;
> The students were unable to understand and use mathematical language and notation;
> The students did not know how to begin proofs. (Moore 1994, p. 252)

Moore's analysis is very rich, and probably contributed to open new directions for research. But such an accumulation of sources may discourage didactical action; trying to develop a teaching design that permits the overcoming of all these problems can seem overambitious!

Moreover, these "sources" have very different natures and can also have different origins, even several intertwined origins. Why do students ignore the definitions? Is it due to a lack of personal effort from each of them, on an individual level? Or is students' personal work inadequate with new university requirements? Are these requirements made explicit by university teachers? I want to emphasize here the importance of the original focus, and attached theoretical frame chosen by the researcher. According to these, different causes for difficulties will be discussed; even the difficulties observed may differ, and naturally the remedial didactical actions too.

Through the years, transition issues have been researched with various theoretical approaches focusing on different transition aspects and leading to the design of different teaching. However, I did not choose here to organize this survey in order to comment on an historical evolution of research. My aim is indeed more to collect all the different views of transition revealed by different approaches, and to investigate how they contribute to the understanding of secondary–tertiary transition. Proposing categories of transition studies was already done by De Guzman et al. (1998). The authors distinguish between epistemological, cognitive, socio-cultural and didactical perspectives. My work continues in a similar direction, but with a different organization of perspectives and focuses.

Students' difficulties are the central point in the studies about transition, because they can indicate the existence of a transition problem. I use the term "difficulty" here to mean the association of an inadequate mathematical production and of various levels of factors likely to have caused this production. Research has an initial focus, connected with its theoretical approach. Each focus uses specific references and material to build an a priori idea of students' difficulties. The a priori idea naturally leads the researcher to observe different students' difficulties, and to develop different means of didactical action. This process eventually yields a specific view of transition. These three kinds of research outcomes, namely, difficulties observed, didactical action, and view of transition are certainly intertwined, and they also modify the a priori idea of difficulties. However I do not examine here such evolutions within a given focus.

I present in Section 2 works focused on the students' thinking modes and organization of knowledge. The works presented in Section 3 focus on proof, and on mathematical communication issues. I consider in Section 4 an institutional perspective, and more precisely a focus on didactical transposition and didactical contract. Thus the focus enlarges along this article from questioning individual behaviors to more socio-cultural and eventually institutional approaches. I propose in Section 5 a synthesis of the research outcomes, trying to analyze how the views of transition provided by these different focuses combine to outline this complex object.

2 Thinking modes and knowledge's organization

The word "transition" is extensively used in the literature belonging to the Advanced Mathematical Thinking field. Tall (1991) declares: "The move from elementary to AMT involves a significant transition: that from describing to defining, from convincing to proving in a logical manner based on definitions" (p. 20). The word "transition" in this quotation characterizes the move between two modes of thinking; does this move coincide

with the secondary–tertiary transition? This question catches up with the recent debates about AMT (and with some older ones as well). As Selden and Selden (2005) question, does the term "advanced" refer to the mathematics, or the thinking, or both? Presenting the most recent works in the field, they discuss the criteria for "advancedness," and claim that advanced thinking can occur at any level. Many authors agree on the need for advanced thinking when the university mathematical content is tackled. Edwards et al. (2005), define AMT by focusing "on the phenomenon that seems to first occur during a mathematics student's experience in undergraduate mathematics when he or she first begins to deal with abstract concepts and deductive proof" (p. 16). They explicitly term the corresponding period a "transitional period." This does not mean that all the mathematics for undergraduates demands advanced thinking, nor that no advanced thinking occurs at other levels. But some of the transition's difficulties might be due to the need for a shift towards AMT. Edwards et al. (2005) define AMT as "thinking that requires deductive and rigorous reasoning about mathematical notions that are not entirely accessible to us through our five senses" (p. 17). Thus the mathematical notions involved play a central part in this definition. This characterization might be a natural consequence of the involvement of some of the authors in APOS theory and associated teaching designs.

In APOS, the analysis of the mathematical thinking required from the students is grounded in a thorough epistemological analysis of mathematical content and knowledge. APOS authors consider different conceptions: action conception, process conception, object conception and eventually schema, with three stages: intra, inter and trans (see for example Weller et al. 2003). The shift from one conception to another can be considered as several stages of a transition process leading to advanced thinking. The analysis of a given piece of mathematical content provides a genetic decomposition which grounds a precise a priori identification of each stage. This a priori analysis permits a precise observation of students' conceptions. It also intervenes as a central tool in the design of appropriate teaching, likely to lead individual students to build the scheme, and thus to develop advanced thinking. Much of such teaching has been organized, following the ACE teaching cycle (activities, classroom discussion and exercises) and using the ISETL software, and its efficiency has been clearly stated (Weller et al. 2003). I retain here the fundamental role played by the analysis of knowledge; the different stages, and thus the idea of transition decomposed into stages too; and the associated development of appropriate teaching.

Robert (1998), using a more general analysis of the mathematical content taught at the end of secondary school and beginning of university, shows that some of the content tackled at that level is intrinsically difficult, because it involves formalizing, unifying, generalizing and simplifying (FUGS). Meeting this new kind of content is one of the difficulties attached to the secondary–tertiary transition. But Robert also develops a global view of a long-term transition process, with the idea of a continuum, instead of stages. Robert writes: "Considering the official curriculum indicates that the mathematics taught from upper secondary school starts looking like (and this resemblance grows with the school years) 'the expert' mathematics (professional mathematicians), for the mathematical content but also for the expected practices" (p. 141, author's translation).

Here, the reference grounding the analysis of transition is not provided by the mathematical content, but by the mathematician's practices and thinking modes (the mathematician is still considered here on an individual level). According to Robert, the mathematician's knowledge is organized. He (or she) does not consider isolated concepts, but organized networks of concepts and properties. Mathematicians know many familiar situations they can use when confronted by a new problem. This organization of

knowledge can be interpreted as a necessary condition to develop the modes of thinking and reasoning required at university. Instead of successive stages for a given concept, the focus here is on a variety of cognitive resources, on their webbing, and a certain kind of flexibility to select one resource or another.

One characteristic of mathematicians when they solve problems seems to be the ability to combine different reasoning, different thinking modes. One can be used to control the others, or to provide a new direction for research when the others fail. A great amount of research about advanced mathematics observes that students at the beginning of university or the end of secondary school would require, and lack, such flexibility.

For example, Sierpinska (2000) distinguishes between theoretical and practical thinking. Theoretical thinking is characterized by organized systems of concepts, and reflection on the semiotic means of representation. Practical thinking is the opposite: it uses prototypical examples, reasoning based on the logic of action. According to Sierpinska, mathematicians use both modes of thinking: most of the time, in a familiar context they think in practical ways, and use theoretical thinking when they meet a new and difficult problem. Many difficulties encountered by first year students can be interpreted as consequences of practical thinking. She observes for example students working in linear algebra who write statements such as: "linear transformations are rotations, dilations, projections, shears, etc. and their combinations with constant parameters" (p. 225). These students focus on the prototypical examples they know, and do not develop a theoretical understanding of the concept of linear transformation.

In a similar way, Lithner (2000) distinguishes between two types of mathematical reasoning: plausible reasoning meant to guide towards what probably is the truth and based on mathematical properties; and reasoning based on established experiences. Observing first year students, he demonstrates that their task solving strategies are mainly based on established experience, and their control is founded on familiarity. Plausible reasoning happens only on very local, limited sub-tasks; thus it cannot be truly efficient.

Sierpinska and Lithner demonstrate that students do not share with mathematicians the ability to develop two kinds of reasoning, two thinking modes. Practical thinking, and reasoning based on established experiences are both useful in mathematics. But while mathematicians also develop other modes of thinking and reasoning, the novice students remain limited to these. Their unique mode is linked with familiar objects and situations, which are sparse, because of their lack of mathematical experience. The enrichment of their mathematical experience along the years is likely to bring a natural solution to this problem. But this process might require a very long time, and as Robert and Schwartzenberger (1991) write about the transition from elementary to advanced mathematics: "There is a quantitative change: more concepts, less time [...]" (p. 133). Could the lack of flexibility be ameliorated, if its source is the lack of mathematical experience?

In his work, Lithner raises a question about another possible reason for such behavior: does the teaching at university permit first year students to experience plausible reasoning? A similar reason is identified by Battie (2003) in her work about number theory at the end of secondary school in France. Number theory was reintroduced in 2002 in the scientific 12th grade curriculum, in order to promote mathematical reasoning. Battie observes 12th grade students working on arithmetic problems. She considers two reasoning dimensions in arithmetic: an organizing dimension (induction, contrapositive, etc.), and an operating dimension (computations, use of familiar theorems, etc.). Students seem able to develop the two dimensions on familiar tasks. But for difficult problems, they remain limited to familiar treatments inside the operating dimension. Battie demonstrates that one origin of this difficulty is the traditional style of teaching at secondary school. Autonomy is left to

students only for a very limited range of prototypical tasks. She studies the example of the following problem:

Find all the integers x and y, with $0 \leq x \leq 200$, $0 \leq y \leq 200$ *solutions of the following system*:

$$\begin{cases} 17x - 11y = 5 \\ \gcd(x, y) = 5 \end{cases}$$

Battie analyzes an exercise proposed to 12th grade students and issued from this problem. The corresponding text constrains students through intermediate questions. They are first asked to find a particular solution of $17x - 11y = 5$ (E). Their course comprises a technique for this task: because gcd(17, 11)=1, they first look for u and v such that $17u - 11v = 1$, using the algorithm termed "Euclid's algorithm" in the course. Then they multiply u and v by 5 to obtain the particular solution. This is a typical operating dimension, left under students' responsibility. They are asked in a second question to determine the general solution; there is in the course another well-known technique for this task. Afterwards they are asked to study the second condition, proving first that if (x, y) is a solution for (E), then "gcd(x, y)=5" is equivalent to "5 divides x." Finally they determine all the solutions. It is clear that this process drastically reduces the organizing dimension in their work. Battie also observes that students are able to produce original reasoning when an opportunity is created; thus the teaching is clearly responsible for the lack of autonomy.

The works focusing on knowledge organization and different forms of flexibility refer to mathematicians' characteristics (instead of mathematics' characteristics as APOS does). But observing differences between students' and mathematicians' behavior does not explain the reasons for these differences, and neither does it provide means for didactical action. In Battie's and Lithner's work, the lack of experience is naturally mentioned as contributing to cause the phenomena observed. But the way the content is taught at upper secondary school or university is also indicated as responsible for the difficulties observed.

Grounding a teaching design in these works would mean first of all changing the range of tasks usually proposed. The new range should comprise a variety of tasks, allowing development of different solutions, in order to foster a form of students' mathematical autonomy. Another direction of didactical action suggested by these works is linked with the use of different representations, as in Sierpinska's work (2000), with the use of a dynamic geometry environment to teach linear algebra. Working with technological devices such as dynamic geometry environments, but also computer algebra systems (Winsløw 2003) or even interactive online exercises (Bookman and Malone 2003; Cazes et al. 2007) can be helpful in fostering flexibility.

Considering the mathematicians as a reference can be done in several ways. In the works presented in Section 3, the authors also refer to mathematicians, but consider social practices inside the mathematical community.

3 Proofs and mathematical communication

Describing her experience of a transition course she taught in the late 1970s, Epp (2003) writes: "My students seemed to live in a different logical and linguistic world from the one I inhabited..." (p. 886).

Similar metaphors are frequently used to evoke what happens during the secondary–tertiary transition. University is seen as a new world, or at least a new country, with a new

language and new laws that make the novice student feel like a foreigner. These are the laws and the language used by mathematicians. Thus here, mathematicians can still be considered as a reference. The focus is not on the way their knowledge is organized, but on the way they build proofs, on the way they communicate these proofs, and more generally on the mathematical language they use. Naturally reasoning and proving modes are strongly intertwined, thus the evolving focus between the previous section and this one can be considered as a gradual change.

Studies considering proof in many different countries have shown that only a minority of students are able to build consistent proofs at the end of high school. We remember the alarming statements by Moore (1994) mentioned in the first section. Harel and Sowder (1998), classifying university students' schemes of proof, demonstrate that many students judge the validity of a proof by external factors, by criteria based on its superficial form, or referring to an external authority. But appropriate teaching can help them to evolve towards analytical proof schemes, where conviction relies on logical deduction. A positive evolution is also recorded by Segal (2000) for first year students in the UK. Segal distinguishes between conviction and validity, as the private and public aspects of proof. She builds a test with different mathematical arguments for a given statement ("the product of two diagonal matrices is diagonal"). Students must say for each argument if they find it convincing or not, and valid or not. Some of the arguments are empirical justifications, others are deductive justifications (some valid, and others not in both cases). The same test is proposed three times during the year (beginning of the year, beginning of the second semester, end of the year). Segal observes significant evolutions of the students' perception of the validity of an argument. Students who enter university massively display in Segal's study inadequate perceptions of proof. During the first year, they seem to consider more and more deductive proofs, correct or not, as valid. According to Segal, students become gradually more influenced by the special "look and feel" of deductive arguments, a phenomenon consistent with Harel and Sowder's results. But at the same time, students clearly progress in identifying a valid empirical proof. Thus the situation, although grave, is perhaps not as alarming as Moore's statements described.

A cause not mentioned by Moore seems to be gaining more and more attention: the teaching received at university could be at least partly responsible for the difficulties encountered by novice students. According to Dreyfus (1999), college teachers and textbooks, which shape the students' behavior, do not help them to establish "socio-mathematical norms," which means in this context, criteria shared by students and teachers to decide whether a proof is valid or not, what is a satisfactory explanation, etc. In the courses he refers to, giving students the means to learn to construct proofs, and to control their validity does not appear as an explicit aim. These aspects of mathematics seem to be left under the students' responsibility. Variations of rigor in proof can be observed even in university textbooks. Thus the rules are not made clear for students. More recently Durand-Guerrier and Arsac (2003) studied, for example, the case of the (ε, δ) proofs. In textbooks, various situations can be observed, even for an identical mathematical content like the definition of limit. Sometimes it is formally written with quantifiers. Sometimes it is formulated in everyday language. Sometimes the link between ε and δ is made explicit by a notation like δ_ε. Sometimes it remains implicit. There is no clear reference proposed to students.

Durand-Guerrier and Arsac (2003) demonstrate that mathematicians judge the validity of a proof using syntactic and semantic aspects simultaneously. They propose to university teachers a student's production in topology. Teachers identify a mistake, but only because they know that the mathematical result cannot be right. These teachers have no criteria

based on logic to decide the validity of the proof. The simple syntactic rules, like "the negation of a quantified statement is obtained by exchanging the ∀ and ∃ signs" are not sufficient for an efficient use of propositions expressed using logical symbols. A semantic control is required in many cases; for example, producing a counter example is a semantic aspect of proof. Weber and Alcock (2004) also distinguish syntactic and semantic proof productions. And they demonstrate that mathematicians building a proof use both, in particular with a semantic control of syntactic manipulations.

What seems most specific of the "advanced mathematical practices" required at university is the syntactic aspect, and the association of semantic and syntactic knowledge.

Syntactic knowledge is strongly connected with the use of symbols, quantifiers in particular. This particular semiotic aspect represents in most countries a rupture when entering university. Syntactic knowledge, and the associated symbolic language, as Iannone and Nardi (2007) state, is useful to define abstract concepts, to produce formal arguments, to check semantic knowledge. It is the language of advanced mathematics, required to enter the mathematical community, and to communicate inside this community.

We enter here more precisely into a socio-cultural view of transition phenomena. According to Iannone and Nardi (2007), syntactic knowledge is the "genre speech" of the mathematical community. And acquiring this specific language starts at the beginning of university. Novice students quickly notice that their teachers use a specific language; and they attempt to imitate the "genre speech" of university mathematics (Nardi and Ianonne 2005). This attempt leads them to produce texts looking like mathematics, focusing sometimes more on the form – many symbols, usual language brought to a minimum – than on the mathematical meaning.

One of the examples quoted by Nardi and Ianonne (2005) corresponds to the following question:

$$\textit{Let } x \in IR \textit{ have the properties that } x \geq 0 \textit{ and } \forall\, n \in IN,\ x < 1/n. \textit{ What is } x?$$

One student writes in his answer:

$$x \in IR,\ x \geq 0,\ \forall\, n \in IN,\ x < 1/n$$

$$\textit{as } n \to \infty\ x \to 0 \textit{ hence for } \forall\, n \in IN\ x = 0$$

It seems indeed to be a formal game. However, the mathematical meaning can come later. If mathematics at university has to be learned like a language, it is a natural process that the students first try to mimic what they read and hear, and consequently formulate absurd sentences. According to Berger (2004), students learn new mathematical signs as new words in a foreign language. Thus they start developing a "functional" use, based on analogy. The above example can be interpreted as a functional use of the signs associated with the notion of limit. With the appropriate commentaries of the teacher, this functional use can evolve to the official use, associated with a real understanding. A natural evolution must happen, with the help of the teacher. But is there enough time for it? And are teachers really helpful?

Chellougui (2004a, b) studies the use of the quantifiers "for all: ∀" and "there exists: ∃" in the first year of university, in particular in textbooks and teachers' practices. She notices many different kinds of formulations, in the same book or by the same teacher. Sometimes the quantified relation remains implicit, formulated in usual language. Sometimes quantifiers are very strictly used; and sometimes they are incorrectly used as shorthand symbols, even by teachers who do not accept that usage from students. First year university teachers should be exemplary on the use of quantifiers, like a primary school teacher on

handwriting. And they do not seem cautious enough. Quantifiers are only one aspect of syntactic knowledge; nevertheless, the phenomena identified by Chellougui are significant, and reveal perhaps a more general tendency, consistent with Dreyfus' (1999) observations about proof.

What do these works tell us about possible didactical action? For Berger (2004) the functional use of a mathematical sign can evolve to an official use, at least with an appropriate teacher's scaffolding. Chellougui's work tends to indicate that this scaffolding is not a usual teacher's practice. Moreover, there is a real need for improvement of the texts proposed to students by the teachers and of the textbooks to clarify the rules and expectations about syntactic knowledge. This would require some form of university teachers' training, which is still considered a bizarre idea by many mathematicians in all countries. Organizing teaching based on APOS also requires some kind of training. But it seems to be due to a major change in the way the content is presented and to the introduction of the ISETL software. Thus it leads to individual teachers' evolution directed towards innovation, which certainly requires a significant endeavor, but is perhaps more easily accepted than an evolution grounded in a critical view of usual practices. The direction indicated by Durand-Guerrier and Arsac (2003) also implicates such a training. Observing that teachers have no criteria based on logic to decide the validity of a proof, they advocate teaching logic to provide both teachers and students with a common set of rules that dictate whether a proof is valid or not. However, they do not organize and evaluate this idea.

As we already observed at the end of the second section, the difficulties recorded here are at least partially resulting from the way mathematics is taught at university. This means questioning neither the content involved, nor the socio-cultural norms, but the way this content and these norms are taught in secondary school and university institutions. This is explicitly done by the works considered in the next section.

4 Didactical transposition and didactical contract

We saw in the second section that the advanced mathematical thinking reports identify epistemological causes for the students' difficulties: the advanced content seems intrinsically difficult. However, all other studies tend to integrate more and more explicitly the way the knowledge is taught, at secondary school or at university. And the knowledge actually taught stems from a didactical transposition process (Chevallard 1992), depending on the institution. Beyond transposition aspects, the secondary–tertiary transition can certainly be considered with an institutional perspective. A shift between two institutional cultures happens when entering university (Artigue 2004). Questioning this change of cultures can lead researchers to consider precise mathematical content, and develop detailed transposition studies. It can also lead researchers to study more general institutional expectations. Secondary school and university have different didactical contracts (Brousseau 1997), and the shift from one contract to another can generate ruptures. Moreover the discourse about what is expected at university must be thoroughly compared with what is actually assessed.

4.1 Mathematical organizations

The anthropological theory of didactics (Chevallard 1992) shows that a given institution proposes mathematical organizations, or praxeologies. A mathematical organization [T, τ,

θ, Θ] has four components: a type of tasks, T, associated techniques, τ, a technology, θ, which is a discourse explaining and justifying the technique, and a theory, Θ, which is itself a discourse justifying the technology. These organizations can be observed in the official curriculum text, or in the textbooks provided by the institutions.

Transition issues can be studied by focusing on mathematical organizations on different levels as we shall see in the following examples.

Investigating the possible uses of geometry to teach linear algebra (Gueudet 2004, 2006), I studied in particular mathematical organizations corresponding to types of tasks existing both in geometry at secondary school (in the French curriculum) and in linear algebra. Let us consider the example of the type of task (T): "Prove, that a given family of *n* vectors is an orthogonal basis of an *n*-space" (naturally in secondary school the space is the space of geometrical vectors of the plane, or of the space). The technique (τ) is the same in both institutions: computing all the scalar products of each pair of two different vectors to prove they are orthogonal. There is no need to prove afterwards that the vectors are independent (or non-collinear, or non-coplanar in the secondary school vocabulary). This fact is justified at university by the following theorem: "A set of orthonormal vectors is independent;" this theorem is the technology (θ) in the university organization. At secondary school, there is no such discourse. The fact that two orthogonal vectors are not collinear is straightforward; there is no need to produce a discourse to justify it. Other observations indicate that, for similar types of tasks, there is no need for justification in the secondary school geometry, because of spatial evidences, while theorems are required in linear algebra. This is an important rupture; the spatial evidences used at secondary school can even become obstacles when students encounter more general cases where vectors can, for example, be orthogonal to themselves.

Is the technological discourse only lacking in this specific case, or is it a general characteristic of the secondary school institution? Raising this question leads to the consideration of mathematical organizations on a broader level.

Bosch et al. (2004) develop such a use to study secondary–tertiary transition in Spain. Proposing a questionnaire to novice university students, they observe similar difficulties for different mathematical content. For a given type a task, most of the students know a single technique, and are unable to use another one. After applying a technique, they are unable to interpret the result obtained. When they are provided with a mathematical model for a given situation, issued from physics, economy etc., they fail to use this model to deduce information about the situation (interpretation and use of a model correspond to a technological level in the mathematical organization). Analyzing secondary school textbooks, the authors demonstrate that the difficulties of the students correspond to characteristics of the mathematical organizations proposed in these books. For a given type of task, in most books a unique technique is proposed. For example, for the type of task "computing the derivative of a given function," there is one technique for each kind of function: the rational fractions must be derived as quotients, etc. Interpreting the result after applying a technique is not required in most cases; and working with a mathematical model (even a model provided in the text of the exercise) is only done in a minority of exercises (about the derivative, 176 exercises out of 1957, less than 9%). These general characteristics of the textbooks indicate choices of the secondary school institution. The authors interpret it by saying that the mathematical organizations at secondary school are limited: focused on a precise, narrow type of task. They are also rigid (associated with a unique technique), and poorly connected to each other. Thus they cannot be integrated into broader mathematical organizations, characterized by a common

technology, and comprising several articulated types of tasks and techniques. The difficulties of the students are not individual difficulties, but consequences of these institutional practices. The students cannot coordinate themselves the limited mathematical organizations they have met to build a complete organization. University teachers seem to consider that, for the type of tasks encountered at secondary school, broad organizations have been built.

The work by Winsløw (2007) about secondary–tertiary transition in calculus confirms Bosch et al.'s results and complements them by precise analyses of university mathematical organizations. According to Winsløw, two types of transitions occur in the mathematical activity proposed. The first one is a shift from an activity focused on the practico-technical block [T, τ] at secondary school towards an activity involving more complete mathematical organizations. In the second transition, types of tasks involving theoretical objects appear: The theoretical block [θ, Θ] of an already known mathematical organization becomes the practico-technical block of a new mathematical organization. This analysis can be characterized as an institutional interpretation of the phenomena understood in a cognitive perspective as encapsulating a process, in order to perform actions on it.

Winsløw observes that the two types of transitions do not necessarily appear at the beginning of university; they can take place later on; and the innovative teaching, grounded in Winsløw's observations and organized at the University of Copenhagen, Denmark (Grønbæk and Winsløw 2006) concerns a second-year real analysis course. According to the authors, this course clearly constitutes a "transition moment."

Focusing on mathematical organizations yields precise results about the way an institution presents specific mathematical content, but also more general observations about transposition features. However this focus does not permit us to identify all the institutional expectations. For example the expectations in terms of autonomy and flexibility are more difficult to grasp only with the mathematical organization tool. Thus other elements must be taken into account to understand the institutional expectations, and the didactical contract in both institutions.

4.2 A complete institutional transition study: the case of the derivative

Praslon (2000) studies the secondary–tertiary transition for the notion of derivative with an institutional perspective. He notices general institutional phenomena, as follows. There is a diversification of the tasks, with new tasks appearing very often and thus not enough time for the students to acquire routines. Proofs provided during the lectures at university play a new role: they are central in the building of the university mathematical culture, because they indicate methods, and also what requires justification or what does not. Different types of results are established in the course: some results provide useful tools, while others are only intermediate results, stages on a path towards an important theorem, etc. All these observations demonstrate that the general features of the university didactical contract differ from what happens at secondary school.

On a more precise scale, Praslon uses the anthropological approach to study the tasks proposed in secondary school and university textbooks around the notion of derivative. He observes in particular that:

- The secondary school tasks are split into simpler subtasks;
- At secondary school, indications are often provided to help the student, who does not need to develop a real mathematical autonomy;

– The derivative intervenes mostly as a tool, for example to study the variations of a function at secondary school, while it can be studied for itself at university (moreover, both tool and object dimensions can intervene in the same exercise at university);
– More flexibility between frames (Douady 1992) or registers (Duval 2006) is required at university.

All these micro-breaches produce a real gap between the ways the derivative is tackled in both institutions. This gap is not the straightforward gap between an intuitive and a formal analysis. It concerns more general abilities, like autonomy in the face of a complex task, the possibility of building a complex proof with several reasoning steps, and the students' attitude towards the mathematical texts they encounter.

Praslon also observes more precise discrepancies between what is expected by the institution and the students' effective abilities. He organizes problem solving sessions on specific mathematical themes at the very beginning of the first year of university. One of these sessions is dedicated to "strongly increasing" functions. A "strongly increasing function" is an increasing function, whose derivative itself is increasing. At the beginning of the session, students have to look for examples of strongly increasing functions, and for criteria ensuring that a given function is strongly increasing on a given domain. Students propose the example of $x \rightarrow e^x$, which is correct; but they generalize then to $x \rightarrow a^x$, for any value of a in IR. They also mention $x \rightarrow x^3$, because of its graphical representation which seems "very increasing," according to one student interviewed. Students also declare that $x \rightarrow a ln|x-a|$ cannot be strongly increasing "because it is a logarithm." Looking for examples, they refer to well-known functions: a typical reasoning of secondary school students, according to Praslon. The reasoning expected in this case at university is grounded in the study of the variation of the derivative by computing the second derivative.

Praslon studies a priori differences between the secondary school and university mathematical cultures, and their consequences in the students' behaviors. He uses the anthropological theory, complemented by analysis tools like registers, frames, autonomy degrees which permit him to identify the multiple micro-breaches. The same approach is further developed by Bloch and Ghedamsi (2004). They introduce ten didactical variables in order to study transition for the notion of limit. These variables include the tasks, the associated techniques, but also the autonomy required, the number of new notions introduced, the process or object status of the notion, and the type of conversion between semiotic representations. Studying textbooks with these variables, the authors identify several micro-breaches. They put a particular emphasis on graphic representations, which are often used at secondary school and are seldom exploited at university (Maschietto 2001).

Studying transition with an institutional focus can lead to design of innovative teaching at secondary school. Bloch (2003) proposes teaching for grade 12 students likely to introduce them to the specific kind of proofs required in analysis. Schneider (2001) and the AHA (heuristic approach to analysis) project team in Belgium build a complete teaching sequence of secondary school calculus grounded in mathematical modeling.

But secondary school does not aim at preparing students for scientific studies at university. Thus these studies will rather lead to design of university teaching comprising first of all revising activities, with no new knowledge but only aiming at building complete mathematical organizations around secondary school's content. The necessity of such reconstructions was already indicated by Artigue (2001). This teaching should also foster students' flexibility and autonomy; they should aim at facilitating the student's entrance in the new didactical contract.

Dreyfus (1999) writes, about the transition to formal proof: "This implies acquiring new attitudes and conceiving of new tasks: the central question changes from "What is the result?" to "Is it true that...?" Students thus need to develop new and more sophisticated forms of knowledge" (p. 106). This change of attitude towards mathematics is certainly a crucial issue about secondary–tertiary transition, and it can be interpreted in terms of a change of didactical contract. At secondary school, students just have to produce results. At university, they seem to have an increasing responsibility towards the knowledge taught.

Much innovative pedagogy tries to foster this change. Scientific debates (Legrand 2001), questioning of mathematical texts (Sackur et al. 2005), but also naturally the numerous courses where problem solving plays a central part (Santos-Trigo 1998; Schoenfeld 1998; Mamona-Downs and Downs 2004, etc.) indicate this change of attitude as one of their aims.

But what does actually happen in the usual university teaching? Does this new attitude correspond to the actual institutional university expectations? This claim cannot be made without considering the assessments proposed.

4.3 Institutional expectations

Comparing secondary school and university teachers' expectations does not seem to have been researched yet. Thus I shall mention here briefly some research investigating actual university expectations, in order to compare it with the results stemming from the analysis of the mathematical texts proposed to the students when they work with a teacher, in a course or in tutorial sessions.

What do university teachers expect from the novice students? According to Rey et al. (2003), university teachers of all domains lack a "target practice": a clear objective for the teaching they deliver. Castela (2004) also identifies this lack for mathematics teachers at university, and suggests it is likely to lead them to teach mathematics "for themselves", which means targeting mathematicians' practices, even though only a minority of their students are likely to become professional mathematicians. She observes simultaneously, like previous authors (Crawford et al. 1998), that the students' personal work is more focused on learning to mimic strategies. Students seem to aim at learning to imitate the solution of prototypical exercises, more than at developing a real understanding. How can such a behavior, far away from the teacher's expectations, be explained?

Lithner (2003) studies 600 exercises in three calculus textbooks comprising exercises with their correction (thus books used by the students for their personal work). He demonstrates that 70% of these exercises can be solved by identifying a similar situation and mimicking the procedure. Most of the students' personal work being done on such textbooks (Castela 2004), they are likely to learn to develop mimicking strategies.

But the main factor identified by these authors is the type of assessment developed at university. In many countries, it seems that the examination texts are elaborated by the teachers themselves. And the desire to avoid a massive failure leads them to propose exercises similar to the exercises studied during the tutorials. Some of the teachers even write textbooks designed to help the students to prepare for these exams, thus textbooks that will eventually lead students to learn to mimic strategies. Students know that during the exam, it will be sufficient to remember a similar situation and the corresponding technique. Thus understanding this technique is superfluous. In France, first year university students take examinations proposed by their own teachers every twelve weeks or less (thus on a very limited content). Comparing them with preparatory classes' students, who work two

years in a row without examinations and then take different national competitions to enter engineering colleges, Castela (2004) observes that the first kind of students develop superficial work, in order to learn to mimic techniques. The second kind of students develop abilities to tackle new problems, and to transfer well-known methods to new situations.

Thus the assessment mode seems to prevent the necessary change of attitude indicated by Dreyfus (1999); it is more likely to lead students to develop an attitude opposed to university teachers' official expectations. Observing assessments ill-aligned to expectations led several research groups to propose and evaluate alternative modes of assessment at university (Ridgway et al. 2001; Haines and Houston 2001); and many innovative teaching designs propose new assessment modes.

I already mentioned the teaching organized at the University of Copenhagen, Denmark (Grønbæk and Winsløw 2006). The usual course comprises lectures, problem solving sessions, a final assessment with a 3-h long written examination, and an oral examination lasting half an hour. The authors added thematic projects to the course. These projects are written assignments centered on a mathematical theme, like: *the Hilbert space* L^2; they comprise several tasks, including open and theoretical tasks; the students work on the projects in groups. A part of the usual problem solving sessions is dedicated to the project, so the groups can ask the teacher for help. An individual student has to work on six projects during the semester. During the oral examination, the student must present one of the six projects (randomly chosen by the student). This experiment led to a significant improvement of the students' achievement and satisfaction. Students' projects have also been the core of assessment for many years in the Warwick project (Alcock and Simpson 2001; Gray et al. 1999; Yosof and Tall 1999), and in the Roskilde University problem-solving oriented projects (Niss 2001).

Nevertheless the question of assessment with a specific secondary–tertiary transition focus has not been studied yet.

5 Synthesis/Conclusion

5.1 Views of transition

As announced in the introduction, I have chosen here to present research studies and their outcomes according to their initial focus and associated theoretical frames. The focuses successively considered were on thinking modes, organization of knowledge, proof and mathematical communication, didactical transposition and didactical contract.

The research belonging to AMT is focused on thinking modes. It refers to mathematical content to build an a priori idea of students' difficulties. Using this a priori idea, it reveals students' difficulties formulated in terms of stages of knowledge that the students do not reach. It permits the building of corresponding teaching designs, presenting knowledge in order to move along the successive stages. Thus according to these works, there is a specific transition for each relevant piece of mathematical content, and this transition is a move along identified stages. The use of the ACE teaching cycle suggests nevertheless that the authors also pay attention to other issues, outside of their original focus.

The research focusing on organization of knowledge and reasoning modes uses the mathematician's organization of knowledge and reasoning modes as reference to build its a priori idea of students' difficulties. The difficulties observed were formulated in terms of students being unable to tackle complex problems, and remaining limited to technical treatments. The means of action suggested, in addition to waiting for the students to build

their own mathematical experience, are directed towards the proposition of a variety of tasks, leaving enough mathematical autonomy. The corresponding view of transition is a long-lasting process, leading to the construction and webbing of different kinds of knowledge and reasoning modes.

The research focusing on proof and language issues refers to practices existing inside the mathematicians' community. The difficulties observed are mathematical productions with an incorrect use of signs (symbols or words), or incorrect syntax; or incoherence in the global articulation of a text. The means of action suggested are mostly turned towards an increasing caution in the treatment of the mathematical texts proposed to students when they enter university. The view of transition offered by these works combines a deep local gap and a long-term evolution. The students indeed have to get progressively accustomed to a new language. But this new language seems to be very different from everything they know, in particular because it uses new signs, and because the "grammatical" rules and the reasoning seem to be more and more intertwined. In fact entering university also means learning new rules, which are not made explicit; thus the transition is also a long enculturation process.

The research focusing on didactical transposition uses university and secondary school textbooks and official curriculum documents to establish an a priori analysis of the students' difficulties. The difficulties observed are usages of an inadequate secondary school technique for a task given at university; but also a lack of autonomy and flexibility. The means of action are directed towards the construction or reconstruction of complete mathematical organizations, and the proposal of challenging tasks. The view of transition offered consists in a number of ruptures, which combine and result in a deep gap.

Didactical transposition and didactical contract issues are strongly intertwined. The official expectations at university emphasize an expected change in the students' attitudes, which is certainly a central factor of the gap mentioned above. But the assessments, whose contribution to the actual expectations is crucial, seem to go in the opposite direction. Thus the deep transition gap is perhaps more a gap between actual students and dream students than between actual students and students succeeding in the examinations.

5.2 Conclusion

The central point I want to emphasize after this survey is the way all these reports of research help to clarify the situation. Secondary–tertiary transition first appeared as a bundle of novice students' problems, resulting from an intricate cluster of causes. The various research contributions on the subject with their different focuses permitted a distinction among individual, social, and institutional phenomena. And they also allowed us to sort out different kinds of difficulties, eventually leading to different views of transition.

Contrasting and combining these different views outlines an object comprising both ruptures and long-term evolutions. Ruptures are observed inside the mathematical concepts tackled; in the introduction of new semiotic objects; but also in the tasks proposed, and in the associated mathematical organizations. At the same time, a more gradual process takes place, with an enrichment of the mathematical resources, their necessary networking, and the flexibility required for an adequate usage of the resources network.

This clarification of the causes plays a fundamental role in the building of appropriate didactical actions. Browsing various innovative teaching designs, we have seen that in addition to the choices driven by research, they share more general pedagogical common features. This indicates the need for further research, in order to make precise the roles and consequences of these features. I want to emphasize here two examples.

The first example is the role of technology. Most of the experimental teaching designs involve various technological tools: ISETL, graphing tools, etc. But the question of the effective and possible uses of technology in the secondary–tertiary transition has not been researched yet, as far as I know. Are the abilities with technology built at secondary school exploited at university? Could technology be helpful to foster novice students' autonomy, for example by using appropriate online resources?

A second important aspect concerns assessment and its consequences for the students' private work. Is the private work required from students really different in upper secondary school and at university? Which are the actual differences between secondary school and university assessments? Is it possible to introduce new assessment modes at university without worsening the rupture with secondary school?

Beyond the need for further research, and for teaching designs grounded in their findings, an important challenge is to disseminate the results obtained. This is probably a delicate issue for all the research addressing tertiary level, in particular because of the lack of teacher training. But secondary–tertiary transition is recognized as problematic by many mathematicians, and thus this issue is likely to provide means to raise their interest in educational issues.

References

Alcock, L., & Simpson, A. (2001). The Warwick analysis project: Practice and theory, In D. Holton (Ed.), *The teaching and learning of mathematics at university level* (pp. 99–112). Dordrecht: Kluwer.

Artigue, M. (2001). What can we learn from educational research at the university level? In D. Holton (Ed.), *The teaching and learning of mathematics at university level* (pp. 207–220). Dordrecht: Kluwer.

Artigue, M. (2004). Le défi de la transition secondaire-supérieur. Que peuvent nous apporter les recherches en didactique des mathématiques? (The secondary–tertiary challenge. What do we learn from mathematics didactics research?) *Paper presented at the first French-Canadian Congress of Mathematical Sciences*, Toulouse.

Battie, V. (2003). *Spécificités et potentialités de l'arithmétique élémentaire pour l'apprentissage du raisonnement mathématique* (*Specificity and potential of elementary number theory for the learning of mathematical reasoning*), Thèse de doctorat de l'Université Paris VII.

Berger, M. (2004). The functional use of a mathematical sign. *Educational Studies in Mathematics, 55,* 81–102.

Bloch, I. (2003). Teaching functions in a graphic milieu: What forms of knowledge enable students to conjecture and prove. *Educational Studies in Mathematics, 52*, 3–28.

Bloch, I., & Ghedamsi, I. (2004). The teaching of calculus at the transition between upper secondary school and university. *Paper presented at the 10th International Congress on Mathematical Education* (ICME10), Copenhague, Denmark.

Bookman, J., & Malone, D. (2003). The nature of learning in interactive technological environments. A proposal for a research agenda based on grounded theory. *Research in Collegiate Mathematics Education V,* 182–204.

Bosch, M., Fonseca, C., & Gascon, J. (2004). Incompletud de las organizaciones matematicas locales en las instituciones escolares (Incompleteness of the mathematical organizations in the educational institutions). *Recherches en Didactique des Mathématiques, 24*(2–3), 205–250.

Brousseau, G. (1997). *Theory of didactical situations in mathematics 1970–1990.* Dordrecht: Kluwer.

Castela, C. (2004). Institutions influencing mathematics students' private work: A factor of academic achievement, *Educational Studies in Mathematics, 57*, 33–63.

Cazes, C., Gueudet, G., Hersant, M., & Vandebrouck, F. (2007). Using e-exercise bases in mathematics: Case studies at university. *International Journal of Computers for Mathematical Learning, 11*(3), 327–350.

Chellougui, F. (2004a). Articulation between logic, mathematics, and language in mathematical practice, *Paper presented at the 10th International Congress on Mathematical Education* (ICME10), Copenhague Denmark.

Chellougui, F. (2004b). *L'utilisation des quantificateurs universel et existentiel en première année universitaire entre l'explicite et l'implicite* (*Between explicit and implicit use of the "for all" and "exists" quantifiers in the first university year*), Thèse de doctorat, Université Claude Bernard Lyon 1 et Université de Tunis.

Chevallard, Y. (1992). Concepts fondamentaux de la didactique: Perspectives apportées par une approche anthropologique (Fundamental concepts of didactics: Perspectives opened by an anthropological approach). *Recherches en Didactique des Mathématiques, 12*(1), 77–111.

Crawford, K., Gordon, S., Nicholas, J., & Prosser, M. (1998). Qualitatively different experiences of learning mathematics at university. *Learning and Instruction, 8*(5), 455–468.

De Guzman, M., Hodgson, B. R., Robert, A., & Villani, V. (1998). Difficulties in the passage from secondary to tertiary education, *Proceedings of the International Congress of Mathematicians,* Berlin, Documenta mathematica, extra volume ICM 1998, 747–762.

Douady, R. (1992). Des apports de la didactique des mathématiques à l'enseignement (Contribution of mathematics didactics to teaching). *Repères Irem, 6,* 132–158.

Dreyfus, T. (1999). Why Johnny can't prove. *Educational Studies in Mathematics,* 38, 85–109.

Durand-Guerrier, V., & Arsac, G. (2003). Méthodes de raisonnement et leurs modélisations logiques. Spécificité de l'analyse. Quelles implications didactiques? (Reasoning methods and their logical models. Specificity of calculus. Which didactical implications?). *Recherches en Didactique des Mathématiques, 23*(3), 295–342.

Duval, R. (2006). A cognitive analysis of problems of comprehension in the learning of mathematics. *Educational Studies in Mathematics, 61*, 103–131.

Edwards, B. E., Dubinsky, E., & Mc Donald, M. A. (2005). Advanced mathematical thinking. *Mathematical Thinking and Learning, 7*(1), 15–25.

Epp, S. (2003). The role of logic in teaching proof. *The Mathematical Association of America Monthly,* 886–899.

Gray, E., Pinto, M., Pitta, D., & Tall, D. (1999). Knowledge construction and diverging thinking in elementary and advanced mathematics. *Educational Studies in Mathematics, 38*, 111–133.

Grønbæk, N., & Winsløw, C. (2006). Developing and assessing specific competencies in a first course on real analysis. *Research in Collegiate Mathematics Education VI*, 99–138.

Gueudet, G. (2004). Rôle du géométrique dans l'enseignement de l'algèbre linéaire (Resorting to geometrical aspects in the teaching of linear algebra). *Recherches en Didactique des Mathématiques, 24*(1), 81–114.

Gueudet, G. (2006). Using geometry to teach and learn linear algebra. *Research in Collegiate Mathematics Education VI*, 171–195.

Haines, C., & Houston, K. (2001). Assessing student project work, In D. Holton (Ed.), *The teaching and learning of mathematics at university level* (pp. 431–442). Dordrecht: Kluwer.

Harel, G., & Sowder, L. (1998) Students' proof schemes: Results from exploratory study, *Research in Collegiate Mathematics Education II*, 234–283.

Iannone, P., & Nardi, E. (2008) The interplay between syntactic and semantic knowledge in proof production: Mathematicians' perspectives, *Proceedings of the Fifth Congress of the European Society for Research in Mathematics Education* (CERME 5), Larnaca, Cyprus (in press).

Legrand, M. (2001). Scientific debate in mathematics courses, In D. Holton (Ed.), *The teaching and learning of mathematics at university level* (pp. 127–136). Dordrecht: Kluwer.

Lithner, J. (2000). Mathematical reasoning in task solving. *Educational Studies in Mathematics, 41*, 165–190.

Lithner, J. (2003). Students' mathematical reasoning in university textbook exercises. *Educational Studies in Mathematics, 52*, 29–55.

Mamona-Downs, J., & Downs, M. (2004). Realisation of techniques in problem solving: The construction of bijections for enumerations tasks. *Educational Studies in Mathematics, 56*, 235–253.

Maschietto, M. (2001). Fonctionnalités des représentations graphiques dans la résolution de problèmes d'analyse à l'université (Functionality of graphical representations for problem solving at university). *Recherches en Didactique des Mathématiques, 21*(1.2), 123–156.

Moore, R. C. (1994). Making the transition to formal proof, *Educational Studies in Mathematics, 27*, 249–266.

Nardi, E., & Iannone, P. (2005) To appear and to be: Acquiring the « genre speech » of university mathematics. *Proceedings of the Fourth Congress of the European Society for Research in Mathematics Education* (CERME 4), San Feliu de Guixols, Spain.

Niss, M. (2001). University mathematics based on problem-oriented student projects: 25 years of experience with the Roskilde model. In D. Holton (Ed.), *The teaching and learning of mathematics at university level* (pp. 153–164). Dordrecht: Kluwer.

Praslon, F. (2000). *Continuités et ruptures dans la transition Terminale S / DEUG Sciences en analyse. Le cas de la notion de dérivée et son environnement* (Ruptures and continuity in the Terminale S / first scientific university year in calculus. The case of the derivative and its environment), Thèse de doctorat de l'Université Paris 7.

Rey, B., Caffieaux, C., Compere, D., Lamme, A., Persenaire, E., Philippe, J. et al. (2003). Les caractéristiques des savoirs enseignés dans les universités et les hautes écoles (Characteristics of the knowledge taught at universities and colleges), Université Libre de Bruxelles – Service des Sciences de l'Education.

Ridgway, J., Swan, M., & Burkhardt, H. (2001). Assessing mathematical thinking via FLAG. In D. Holton (Ed.), *The teaching and learning of mathematics at university level* (pp. 423–430). Dordrecht: Kluwer.

Robert, A. (1998). Outils d'analyse des contenus mathématiques à enseigner au lycée et à l'université (Tools for the analysis of the mathematical content taught at high school and university). *Recherches en Didactique des Mathématiques, 18*(2), 139–190.

Robert, A., & Schwartzenberger, R. (1991). Research in teaching and learning mathematics at an advanced level. In D. Tall (Ed.), *Advanced mathematical thinking* (pp. 127–139). Dordrecht: Kluwer.

Sackur, C., Assude, T., Maurel, M., Drouhard, J.-P., & Paquelier, Y. (2005). L'expérience de la nécessité épistémique (Experiencing the intrinsic necessity). *Recherches en Didactique des Mathématiques, 25*(1), 57–90.

Santos-Trigo, M. (1998). On the implementation of mathematical problem solving instruction: Qualities of some learning activities. *Research in Collegiate Mathematics Education III,* 71–80.

Schneider, M. (2001). Praxéologies didactiques et praxéologies mathématiques. A propos d'un enseignement des limites au secondaire (Didactic and mathematical praxeologies. A teaching design for limits at secondary school). *Recherches en Didactique des Mathématiques, 21*(1.2), 7–56.

Schoenfeld, A. H. (1998). Reflections on a course in mathematical problem solving. *Research in Collegiate Mathematics Education III,* 81–113.

Segal, J. (2000). Learning about mathematical proof: Conviction and validity. *Journal of Mathematical Behavior, 18*(2), 191–210.

Selden, A., & Selden, J. (2005). Perspectives on advanced mathematical thinking. *Mathematical Thinking and Learning, 7*(1), 1–13.

Sierpinska, A. (2000). On some aspects of students' thinking in linear algebra, In J.-L. Dorier (Ed.), *On the teaching of linear algebra* (pp. 209–246). Dordrecht: Kluwer.

Straesser, R. (2008). A propos de la transition du secondaire vers le monde du travail (About transition from secondary school to the workplace), In A. Rouchier, et al. (Eds.), *Actes de la XIIIième école d'été de didactique des mathématiques*, La Pensée Sauvage Editions Grenoble France (in press).

Tall, D. (1991). *Advanced Mathematical Thinking.* Kluwer, Dordrecht.

Weber, K., & Alcock, L. (2004). Semantic and syntactic proof productions. *Educational Studies in Mathematics, 56*, 209–234.

Weller, K., Clark, J., Dubinsky, E., Loch, S., McDonald, M., & Merkovsky, R. (2003). Students performance and attitudes in courses based on APOS theory and the ACE teaching cycle. *Research in Collegiate Mathematics Education V,* 97–131.

Winsløw, C. (2003). Semiotic and discursive variables in CAS-based didactical engineering. *Educational Studies in Mathematics, 52*, 271–288.

Winsløw, C. (2008). Transformer la théorie en tâches: La transition du concret à l'abstrait en analyse réelle (Turning theory into tasks: Transition from concrete to abstract in calculus), In Rouchier, A. et al. (Eds.), *Actes de la XIIIième école d'été de didactique des mathématiques*, La Pensée Sauvage Editions Grenoble France (in press).

Wood, L. (2001). The secondary–tertiary interface. In D. Holton (Ed.), *The teaching and learning of mathematics at university level* (pp. 87–98). Dordrecht: Kluwer Academic Publishers.

Yosof, Y. M., & Tall, D. (1999). Changing attitudes to university mathematics through problem solving. *Educational Studies in Mathematics, 37*, 67–82.